

# Mathematics - Probability and Statistics

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# Syllabus

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- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
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# Conditional Probability & Independence

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## Session 6

# Conditional Probability & Independence

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## Session 6

### **Objectives:**

- Learn how to determine if two events are independent.
- Understand the definition of conditional probability.
- Learn how to find the probability of the intersection of two events if the two events are independent.
- Learn how to determine if three or more events are pairwise independent.
- Learn how to determine if three or more events are mutually independent.

# WHY?

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**This is one of the most fundamental concepts in probability theory.**

**What happens if you obtain additional information, how should you update probabilities of events?**

For example,

- What's the chance to draw a queen in a 52 card-deck?
  - $4/52=7.7\%$
- **Now, I give you an additional information.** You card is a face, what is now the new probability?
  - **$1/3=33\%$**

# Conditional probability

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The conditional probability of an event A, given an event B with  $P(B) > 0$ , is defined

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Therefore, conditioning **narrows** the sample space to the outcomes consistent with what we know

What's the chance to draw a queen in a 52 card-deck if I know it is a face card?

A=Having a queen

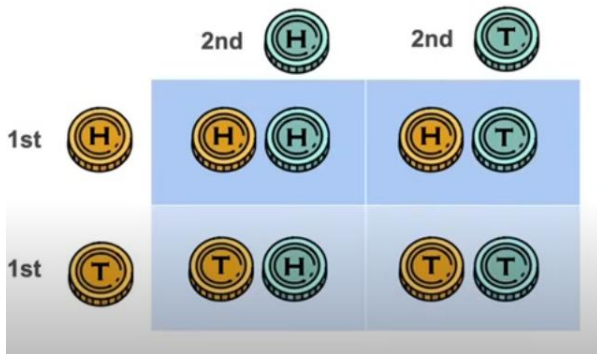
B=It is a face  $\rightarrow P(B)=12/52=3/13$

$P(A \cap B)$ =a card that is both a Queen and a face card. Since all Queens are face cards  $= P(A)=4/52=1/13$

$P(A|B)=1/13 / (3/13)=1/3$

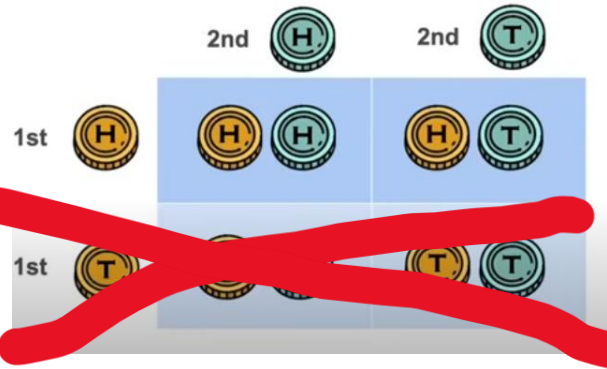
# Conditional probability

**Example: Probability of having 2 head if the first toss is a head?**



- $n_A=1$
  - $N_{\text{tot}}=4$
- $P=1/4$

**condition**



- $n_A=1$
  - $N_{\text{tot}}=2$
- $P=1/2$

## Check with formula

$P(B)$ =proba that first toss is a head=  $\frac{1}{2}$

$P(A \cap B)$  if both are heads, the first is automatically a head, so this is just the probability of two heads=  $\frac{1}{4}$

$$P(A|B) = P(A \cap B) / P(B) = \frac{1/4}{1/2} = \frac{2}{4} = \frac{1}{2}$$

# Conditional probability

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**Exercise:** A family has two children.

- What is the probability that both are boys given that **at least one is a boy**?
- What is the probability that both are boys given that the **older child is a boy**?



# Conditional probability

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**Exercise:** A family has two children.

- What is the probability that both are boys given that **at least one is a boy**?

**Sol:**

- Event A= BB
- Event B= At least one is a boy=3/4 (BB,BG,GB)
- $P(A \cap B) = 1/4$

$$P(A|B) = P(A \cap B) / P(B) = (1/4) / (3/4) = 1/3$$

- What is the probability that both are boys given that the **older child is a boy**?

**Sol:**

- Event A= BB
- Event B= At least one is a boy=(BB,BG)/(BB,BG,GB,GG)=1/2
- $P(A \cap B) = 1/4$

$$P(A|B) = P(A \cap B) / P(B) = (1/4) / (1/2) = 2/4 = 1/2$$

# Conditional probability

**Exercise:** Using the following table, please derive:

- $P(A)$
- $P(B)$
- $P(A|B)$
- $P(B|A)$
- $P(B|A^c)$

|            | Flu | No Flu | Total |
|------------|-----|--------|-------|
| Smoker     | 60  | 140    | 200   |
| Non-smoker | 120 | 680    | 800   |
| Total      | 180 | 820    | 1000  |

- Is flu independent of smoking?

Where:

- $A = \text{Smoker}$
- $B = \text{Flu}$

# Conditional probability

**Exercise:** Using the following table, please derive:

**Sol:**

**A=Smoker**

**B=Flu**

|            | Flu | No Flu | Total |
|------------|-----|--------|-------|
| Smoker     | 60  | 140    | 200   |
| Non-smoker | 120 | 680    | 800   |
| Total      | 180 | 820    | 1000  |

- $P(A)=(200/1000)=20\%$
- $P(B)=(180/1000)=18\%$
- $P(A|B)=P(A \cap B)/P(B)=(60/1000)/(180/1000)=60/180=33\%$
- $P(B|A)=P(B \cap A)/P(A)=(60/1000)/(200/1000)=60/200=30\%$
- $P(B|A^c)=P(B \cap A^c)/P(A^c)=(120/1000)/(800/1000)=120/800=15\%$
- Is flu independent of smoking?

No,  $P(B|A)$  and  $P(B)$  are different, therefore, it is dependent

# Conditional probability

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**Exercise:** A medical test for a disease D has outcomes + and –

|          | <b>D</b> | <b>D<sup>c</sup></b> |
|----------|----------|----------------------|
| <b>+</b> | 0.009    | 0.099                |
| <b>-</b> | 0.001    | 0.891                |

- Is the test accurate?

# Conditional probability

**Exercise:** A medical test for a disease D has outcomes + and –

|          | <b>D</b> | <b>Dc</b> |
|----------|----------|-----------|
| <b>+</b> | 0.009    | 0.099     |
| <b>-</b> | 0.001    | 0.891     |

**Sol**

- Is the test accurate? The test should be + if you have the disease, and – otherwise.

$$P(+|D) = P(+ \cap D) / P(D) = 0.009 / (0.009 + 0.001) = 0.009 / 0.010 = 90\%$$

$$P(-|Dc) = P(- \cap Dc) / P(Dc) = 0.891 / (0.099 + 0.891) = 0.891 / 0.99 = 90\%$$

The test is accurate at 90%

# Conditional probability

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**Exercise:** A medical test for a disease D has outcomes + and –

|          | <b>D</b> | <b>Dc</b> |
|----------|----------|-----------|
| <b>+</b> | 0.009    | 0.099     |
| <b>-</b> | 0.001    | 0.891     |

- Suppose you go for a test and get a positive. What is the probability you have the disease?

# Conditional probability

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**Exercise:** A medical test for a disease D has outcomes + and –

|          | <b>D</b> | <b>Dc</b> |
|----------|----------|-----------|
| <b>+</b> | 0.009    | 0.099     |
| <b>-</b> | 0.001    | 0.891     |

**Sol**

- Suppose you go for a test and get a positive. What is the probability you have the disease?

**90% is WRONG!! Because the disease is rare so you need to take this into account.**

$$P(D|+) = P(D \cap +) / P(+)=0.009/(0.009+0.099)=8.3\%$$

# Independence

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In this special case, knowledge that the outcome of the experiment is in event B does not affect the probability that the outcome is in event A, **A and B are independent.**

**Definition: Two events are independent if any one of the following equivalent statements is true:**

- $P(A|B)=P(A)$
- $P(B|A)=P(B)$
- $P(A \cap B) = P(A) \cdot P(B)$

Ex: Let A be the event that it rains tomorrow, and suppose that  $P(A)=1/3$ . Also suppose that I toss a fair coin; let B be the event that it lands heads up. We have  $P(B)=1/2$

Clearly, both event should be independent as tossing a fair coin does not influence the weather:

$$P(A|B)=1/3=P(A)$$



# Independence

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**Exercise: Results from the previous exercise on flu.**

- $P(A) = (200/1000) = 20\%$
- $P(B) = (180/1000) = 18\%$
- $P(A|B) = 33\%$
- $P(B|A) = 30\%$
- $P(B|A^c) = 15\%$
  
- Is flu (B) independent of smoking (A)?

A and B are independent is  $P(A|B) = P(A)$

$P(A|B) = 0.33$ ,  $P(A) = 0.20$

$P(B|A) = 0.3$ ,  $P(B) = 0.18$

# Independence

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## Helpful theorem:

- If  $A$  and  $B$  are independent, therefore,  $A$  and  $B^c$  are also independent
- If  $A$  and  $B$  are independent, therefore,  $A^c$  and  $B$  are also independent
- If  $A$  and  $B$  are independent, therefore,  $A^c$  and  $B^c$  are also independent

# Independence

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## Exercise:

- Demonstrate: If  $A$  and  $B$  are independent, therefore,  $A^c$  and  $B^c$  are also independent

# Independence

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## Solution:

- Demonstrate: If A and B are independent, therefore,  $A^c$  and  $B^c$  are also independent

$$A^c \cap B^c = (A \cup B)^c$$

$$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(A \cap B)]$$

$$= 1 - P(A) - P(B) + P(A \cap B) \text{ if A and B are independent ; } \mathbf{P(A \cap B) = P(A) \cdot P(B)}$$

$$= 1 - P(A) - P(B) + P(A) \cdot P(B)$$

$$= (1 - P(A))(1 - P(B))$$

$$= P(A^c) \cdot P(B^c)$$

# Independence

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## **Exercise:**

72% of the French population loves eating pizza. If two people are randomly selected from the population, what is the probability that the first person loves eating pizza, while the second one does not?

# Independence

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## Solution:

72% of the French population loves eating pizza. If two people are randomly selected from the population, what is the probability that the first person loves eating pizza, while the second one does not?

A=First person loves pizza

B=Second person loves pizza

We want to measure  $P(A \cap B^c)$  and A and B are independent ( randomly selected)

If A and B are independent A and  $B^c$  also.

$$P(A \cap B^c) = P(A) \cdot P(B^c) = 0.72 \cdot 0.28 = 0.202$$

# Independence

When considering three or more events, we can extend the definition of independence:

Three events  $A$ ,  $B$ , and  $C$  are independent if **all** of the following conditions hold

Mutually independent  $\left\{ \begin{array}{l} \text{pairwise independent} \left\{ \begin{array}{l} P(A \cap B) = P(A)P(B), \\ P(A \cap C) = P(A)P(C), \\ P(B \cap C) = P(B)P(C), \\ P(A \cap B \cap C) = P(A)P(B)P(C). \end{array} \right. \end{array} \right.$

# Independence

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**Exercise:** Suppose we toss a **fair coin twice**. Define the events:

A= “First toss is Heads”

B= “Second toss is Heads”

C= “Both tosses are the same”

- Compute the probabilities  $P(A)$ ,  $P(B)$ , and  $P(C)$ .
- Compute pairwise intersections:
  - $P(A \cap B)$
  - $P(A \cap C)$
  - $P(B \cap C)$

Check whether **pairs of events are independent**.

- Compute the triple intersection  $P(A \cap B \cap C)$ . Compare it with  $P(A)P(B)P(C)$
- Conclude: are the events A,B,C **independent in the strong sense** (mutually independent), or only pairwise independent?



# Independence

**Solution:** Suppose we toss a **fair coin twice**. Define the events:

A= “First toss is Heads”

B= “Second toss is Heads”

C= “Both tosses are the same”

Sample space: {HH,HT,TH,TT}, each with probability 1/4.

- **Single probabilities:**

$$P(A)=1/2=P(B)$$

$$P(C)=1/2$$

- **Pairwise intersections:**

$$P(A \cap B)=P(\{HH\})=1/4=P(A)P(B)$$

$$P(A \cap C)=P(\{HH\})=1/4=P(A)P(C)$$

$$P(B \cap C)=P(\{HH\})=1/4=P(B)P(C)$$

**All pairs are independent.**

- **Triple intersection:**

$$P(A \cap B \cap C)=P(\{HH\})=1/4$$

$$P(A)P(B)P(C)=1/8.$$

Not equal  $\Rightarrow$  **not mutually independent.**

# Independence $\neq$ disjoint

One common mistake is to confuse independence and being disjoint. These are completely different concepts.

When two events  $A$  and  $B$  are disjoint it means that if one of them occurs, the other one cannot occur, i.e.,  $A \cap B = \emptyset$ , therefore  $P(A \cap B) = 0$ .

When two events  $A$  and  $B$  are independent,  $P(A \cap B) = P(A) \cdot P(B)$

| Concept     | Meaning                                                             | Formulas                                               |
|-------------|---------------------------------------------------------------------|--------------------------------------------------------|
| Disjoint    | $A$ and $B$ cannot occur at the same time                           | $A \cap B = \emptyset,$<br>$P(A \cup B) = P(A) + P(B)$ |
| Independent | Knowing whether $B$ happened does not change the probability of $A$ | $P(A   B) = P(A),$<br>$P(A \cap B) = P(A)P(B)$         |

# Multiplication Rule

The probability of the intersection of two events is often needed. The conditional probability can be rewritten to provide a formula known as the multiplication rule for probabilities

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A).$$

**For three events:**

$$P(A \cap B \cap C) = P[(A \cap B) \cap C] = \underbrace{P(C|A \cap B)}_a \times \underbrace{P(A \cap B)}_b$$

$$\text{But since } P(A \cap B) = \underbrace{P(B|A) \times P(A)}_b:$$

$$P(A \cap B \cap C) = \underbrace{P(C|A \cap B)}_a \times \underbrace{P(B|A) \times P(A)}_b$$

**For N events:**

$$\mathbf{P}\left(\cap_{i=1}^n A_i\right) = \mathbf{P}(A_1)\mathbf{P}(A_2 | A_1)\mathbf{P}(A_3 | A_1 \cap A_2) \cdots \mathbf{P}\left(A_n | \cap_{i=1}^{n-1} A_i\right).$$

# Multiplication Rule

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**Exercise:** If an aircraft is present in a certain area, a radar correctly registers its presence with probability 0.99. If it is not present, the radar falsely registers an aircraft presence with probability 0.10.

We assume that an aircraft is present with probability 0.05.

- What is the probability of false alarm (a false indication of aircraft presence)?
- What is the probability of missed detection (nothing registers, even though an aircraft is present)?

# Multiplication Rule

**Exercise:** If an aircraft is present in a certain area, a radar correctly registers its presence with probability 0.99. If it is not present, the radar falsely registers an aircraft presence with probability 0.10. We assume that an aircraft is present with probability 0.05.

- **What is the probability of false alarm (a false indication of aircraft presence)?**

**Sol:** Let define

- Event A=the radar register an aircraft present
- Event B= the aircraft is present

Therefore:            Event  $A^c$ = the radar does not detect an aircraft  
                          Event  $B^c$ = the aircraft is not present

What do we know:

- $P(B)=0.05 \rightarrow P(B^c)=0.95$
- $P(A|B)=0.99$
- $P(A|B^c)=0.10$

What we need: “False alarm” = radar registers presence (A) **but no aircraft is present** ( $B^c$ )

$$P(A \cap B^c) = P(A|B^c) * P(B^c) = 0.10 * 0.95 = 9.5\%$$

# Multiplication Rule

**Exercise:** If an aircraft is present in a certain area, a radar correctly registers its presence with probability 0.99. If it is not present, the radar falsely registers an aircraft presence with probability 0.10. We assume that an aircraft is present with probability 0.05.

- **What is the probability of missed detection (nothing registers, even though an aircraft is present)?**

**Sol:** Let define

- Event A=the radar register an aircraft present
- Event B= the aircraft is present

What do we know:

- $P(B)=0.05 \rightarrow P(B^c)=0.95$
- $P(A|B)=0.99$
- $P(A|B^c)=0.10$
- $P(A^c|B)=0.01$

What we need: “Missed detection” = radar shows nothing ( $A^c$ ) **even though an aircraft is present** (B)

$$P(A^c \cap B) = P(A^c | B) * P(B) = 0.01 * 0.05 = 0.05\%$$

# Exercise

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## Exercise:

Paul have to choose a password that contain at least three but not more than five characters. He can use letters (a-z) or Numbers (0-9).

- How many different possibilities are there for this password if each character can only be used once?
- Suppose that a password follow all the rules, what is the probability that this password is an arrangement of his name? (paul,pual,etc)

# Exercise

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## Solution:

Paul have to choose a password that contain at least three but not more than five characters. He can use letters (a-z) or Numbers (0-9).

- How many different possibilities are there for this password if each character can only be used once?

## ORDER MATTERS

$(26+10) \cdot (26+10-1) \cdot (26+10-2) \cdot (26+10-3) \cdot (26+10-4) = 45,239,040$  or **P(36,5)**

$(26+10) \cdot (26+10-1) \cdot (26+10-2) \cdot (26+10-3) = 1,413,720$  or **P(36,4)**

$(26+10) \cdot (26+10-1) \cdot (26+10-2) = 42,840$  or **P(36,3)**

$$P(n, k) = \frac{n!}{(n - k)!}$$

So total of: 46,695,600



# Exercise

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## Solution:

- Suppose that a password follow all the rules, what is the probability that this password is an arrangement of his name?

We have the total number of possibilities: 46,695,600

We need to know how permutation we can do with the name “Paul”

$$P(n, k) = \frac{n!}{(n - k)!}$$

$$P(4,4) = 4!/0! = 24$$

Probability that the password only contains letters (P,A,U,L) is 24/ 46,695,600

# Exercise

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**Exercise:** Three cards are drawn from an ordinary 52-card deck without Replacement.

What is the probability that none of the three cards is a heart.

# Exercise

**Solution:** Three cards are drawn from an ordinary 52-card deck without Replacement.

What is the probability that none of the three cards is a heart.

$A_1$ =first card is not a heart

$A_2$ =second card is not a heart

$A_3$ =third card is not a heart

$$\mathbf{P}\left(\cap_{i=1}^n A_i\right) = \mathbf{P}(A_1)\mathbf{P}(A_2 | A_1)\mathbf{P}(A_3 | A_1 \cap A_2) \cdots \mathbf{P}(A_n | \cap_{i=1}^{n-1} A_i).$$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) * P(A_2 | A_1) * P(A_3 | A_1 \cap A_2)$$

$$P(A_1) = 39/52$$

Given that the first card is not a heart, we are left with 51 cards, 38 of which are not hearts

$$P(A_2 | A_1) = 38/51$$

Given that the first and second cards are not a heart, we are left with 50 cards, 37 of which are not hearts

$$P(A_3 | A_1 \cap A_2) = 37/50$$

